

AUTHBC: Co-Optimizing Encoding, Authentication Placement, Signature Scheme, and Batching for Blockchain-Grade UAV Telemetry over 802.11

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AUTHBC Project — Thesis draft (auto-generated from frozen results)

Abstract—Flying ad-hoc networks (FANETs) stream tiny telemetry records that must be individually authenticated and hash-chained into a tamper-evident ledger. Because a cryptographic signature (64–96 B) rivals or exceeds the payload (45–190 B), naive per-record signing wastes airtime, energy, and goodput on a shared, lossy 802.11 channel. We argue that four design knobs—record *encoding*, authentication *placement*, signature *scheme*, and *batch* size—are coupled and must be *co-optimized*. We formalize this with five results (overhead dominance, batching amplification, a loss-robustness frontier, a power-independent scheme crossover, and the co-design optimum), and we distinguish these from two boundaries that are *applications of established results rather than new ones*: the regime in which batching amplification ceases to apply, which is the standard diminishing-returns behaviour of packet aggregation, and a payload-exclusion bound [1] that delimits where co-design is possible at all. We implement measurable models for each, and test one falsifiable criterion: the co-optimized configuration must cut on-air authentication bytes by $\geq 40\%$ versus an inline-signature CBOR baseline while keeping verifiability $V_{\text{tgt}} \geq 0.95$ under an assumed per-frame loss $p = 0.05$. We report that criterion as pre-registered even though our analysis later showed the authentication ratio to be a weak headline (Sec. V); moving the goalposts after seeing the data would be the larger error. We also state plainly that the two halves were not equally at risk: for the frame-level placements $V_{\text{tgt}} = 1 - p$ independently of encoding, batch and scheme, so with p fixed the verifiability half is satisfied by construction and the *byte reduction carried the falsifiability*. Measured verifiability, written V_{meas} throughout, is a separate quantity obtained from simulation. Driven by measured microbenchmarks, the optimizer selects a delta encoding under self-batch placement, cutting *total* on-air bytes by 58.7% (to 72.0 B/record) at $V = 0.95$ —criterion met. We report this as a decomposition rather than a single figure, because the axes do not contribute equally: placement and batching cut *authentication* bytes by 75.0% (108→27 B/record) and encoding cuts *payload* bytes by 32.1%, contributing 79% and 21% of the saving respectively. The authentication term alone reduces to $1 - 1/b$, invariant to header size, signature size and encoding—so the co-design claim rests not on that ratio but on *joint feasibility*: coupling the byte model to an NS-3-validated broadcast channel shows the co-design carrying between $1.9\times$ and $3.3\times$ the swarm of an inline baseline on the same medium, depending on the operating point and on whether the limit is read at saturation throughput or at the measured verifiability boundary. We report the operating region rather than a point, and adopt as primary the configuration that complies with 3GPP TS 22.125 (50 Hz, 100 ms, the PX4 companion stream rate): because the batch obeys $b \leq \Lambda D_{\text{max}}$, compliance is free in bytes and costs swarm size, $N \leq 213$ falling to $N \leq 100$. It is not free everywhere—holding $b=4$ under a 100 ms deadline requires $\Lambda \geq 40$ Hz, below which the deadline forces $b=1$ and the saving collapses to

12.2%. Freshness, not the MTU, is what binds: the byte-optimal batch cuts 96.7% of authentication but incurs 1.50 s of staleness, so a co-design optimizer blind to freshness solves the wrong problem. The airtime model is validated against NS-3 3.48 over 30 seeds: unicast matches Bianchi within $+1.29/-0.40\%$, while the common reduction of that model to broadcast fails by $16\times$ at $N=50$ and is replaced by Ma and Chen’s broadcast model [2], [3], whose goodput our measurements confirm to within 0.51% (the success-probability figure is looser, 2.49%). The band is quoted at the 1400 B frame it was measured on: at the 72 B end of the operating range the broadcast model tightens further, to $\pm 0.21\%$, while unicast carries a systematic -1.4 to -2.6% bias that we report rather than absorb. All results are reproduced deterministically from frozen data by an automated staleness gate. AUTHBC introduces no new primitive; its contribution is the rigorous, reproducible co-design and its quantified trade-offs.

Index Terms—FANET, UAV telemetry, authentication, signature aggregation, blockchain, CBOR, 802.11, Bianchi model.

I. INTRODUCTION

A fleet of UAVs streams telemetry—position, velocity, altitude, battery, mode—at tens of records per second per node over 802.11 into a blockchain-grade, hash-chained, signed ledger. We adopt $\Lambda = 50$ rec/s with a $D_{\text{max}} = 100$ ms freshness bound as the primary operating point: it is PX4’s MAVLINK_MODE_ONBOARD companion stream rate and it satisfies 3GPP TS 22.125 [4], and we report the surrounding region rather than defend a point (Sec. V). Each record must be *authenticated* (a signature so a verifier trusts its origin) and *chained* (a hash of the previous record, so the log is tamper-evident).

The difficulty is scale mismatch. Telemetry records are tens of bytes; an Ed25519/ECDSA signature is 64 B, a SHA-256 chain hash is 32 B, and a BLS signature is 96 B. Signing every record inline makes authentication overhead rival or exceed the payload, and on a shared, contended, lossy channel that wasted airtime directly costs goodput, energy, and freshness. How much overhead one actually pays depends on four *coupled* choices: the record *encoding* e (setting the size s), the authentication *placement* (inline, self-batch, relay-aggregate, or block-level), the signature *scheme* σ , and the *batch* size b . Our thesis is that these knobs are not separable and that co-optimizing them jointly beats tuning any one. We make this precise (Sec. III), implement it (Sec. IV), and test the falsifiable headline of Sec. V.

II. RELATED WORK AND POSITIONING

Citation integrity. Every reference in this section was confirmed against the Crossref record for its DOI, and no figure is quoted from a source we have not read. Where a source establishes prior art for something we might otherwise have claimed, that is said explicitly rather than left to the reader.

(a) **Signatures for constrained authentication.** ECDSA [5] yields 64 B P-256 signatures at 128-bit security but is randomness-sensitive; Ed25519 [6], [7] uses a twisted-Edwards curve with deterministic nonces for fast, misuse-resistant, batchable verification; BLS [8] gives the shortest signatures via pairings, and [9] aggregates n signatures into one at the cost of expensive pairing verification. AUTHBC introduces *no new primitive*; it quantifies *where and when* each scheme wins once placement and batching are fixed (Sec. III-D), finding Ed25519 dominant for self-batch and BLS worthwhile only for cross-signer relay traffic.

(b) **Batch/aggregate verification in vehicular/IoT networks.** Zhang *et al.* [10] let a roadside unit “verify multiple received signatures at the same time”, motivated by exactly the pressure we face from the other side: a hard deadline, in their case DSRC’s 300 ms, within which sequential verification does not scale. Their batching is *receiver-side* and buys verification time; ours is *sender-side* and buys airtime, and the two are complementary rather than competing. AUTHBC generalizes from receiver batching to authentication *placement*, co-designs it with encoding and scheme, and adds the loss-robustness frontier that pure aggregation ignores.

(c) **Blockchain/hash-chained ledgers for UAV/IoT provenance.** Nakamoto’s hash chain [11] makes a log tamper-evident; UAV-blockchain systems, surveyed in [12], place telemetry or attestations on such a chain, gaining integrity at per-record overhead cost. AUTHBC targets exactly that substrate byte/energy cost and is agnostic to the consensus layer.

(d) **Compact serialization and delta coding.** CBOR [13] and MessagePack [14] give compact binary encodings; delta coding of telemetry time series is established practice, most prominently in Gorilla [15], which stores delta-of-delta timestamps and XOR-folded values for a $10\times$ reduction. Our variant differs in one respect that the link forces: Gorilla is a database and can chain deltas indefinitely, whereas a lossy broadcast link makes an unbounded chain fragile, so we re-anchor on a keyframe every K records and pay the resulting size for loss independence. AUTHBC measures these head-to-head and shows encoding is coupled to authentication: a smaller payload raises the auth fraction, which *increases* the value of batching.

(e) **Analytical 802.11 modelling.** Bianchi [16] models the DCF as a per-station Markov chain whose fixed point yields closed-form saturation throughput. AUTHBC uses it as the airtime cost model inside the optimizer and validates it against NS-3 [17].

(f) **Broadcast authentication for UAS, and why it is not a baseline.** The closest published system is TBRD [18], a TESLA-based [19] authenticator for UAS Remote ID broadcast, reporting $\approx 50\%$ lower authentication overhead than

TABLE I

TWO WAYS TO BUY AUTHENTICATION BYTES WITH LATENCY. NOT COMPETING DESIGNS: THE LAST ROW IS A HARD REQUIREMENT FOR A LEDGER, AND IT IS WHAT RULES TESLA OUT HERE.

	TESLA / TBRD [18]	AUTHBC (this work)
auth bytes/msg	$\approx 50\%$ of a signature	$(H_f + g_a)/b = 27$ B at $b=4$
what is delayed	verification (key disclosure)	transmission (batch fill)
where the delay sits	at the receiver	at the sender
verifiable on arrival	no — awaits the next interval	yes — each frame self-verifies
loss of one unit	breaks a whole key interval	loses one frame, $V = 1 - p$
non-repudiation	no (symmetric MAC)	yes (signature)

digital signatures. It is *not* a comparator for this work, and the reason is instructive rather than incidental: TESLA authenticates with a MAC and discloses the key later, so it is symmetric and provides *no non-repudiation*. A provenance ledger needs a record to remain attributable to its originator after the fact, which a disclosed key cannot deliver. The two systems are therefore not substitutes, and a byte-for-byte overhead comparison between them would be misleading.

What makes the pairing worth stating is that both buy authentication bytes with the *same* currency — latency — but spend it at opposite ends of the link (Table I).

TESLA is preferable where staleness of *data* is intolerable but staleness of *trust* is acceptable; the reverse holds for a tamper-evident ledger, where a record that cannot yet be verified cannot yet be committed. We claim no superiority over TBRD. What our analysis contributes to that comparison is that the byte saving from batching is exactly $1 - 1/b$ and hence bounded by the freshness budget — a constraint TESLA sidesteps by moving the delay to the receiver, and one that any future hybrid would have to confront.

Overall. AUTHBC is a rigorous, hardware-validated co-design and measurement study, not a new construction. Prior work typically fixes three of the four knobs; we jointly optimize all four under a single falsifiable criterion with a reproducibility gate. Where the vehicular literature builds *new* aggregate-signature constructions to cut broadcast overhead, we ask a complementary question: which combination of *existing, standardised* primitives and system parameters is feasible on a given link, and where is no combination feasible at all.

III. SYSTEM MODEL AND THEORY

A record is integer-only fixed-point (floats break canonical-CBOR determinism) with a 32 B `prev_hash` chain link; its encoded size is s . A frame carries b records, a header $H_f \approx 40$ B, and an authentication object g_a (64 B Ed25519/ECDSA, 96 B BLS), bounded by the MTU $M = 1500$ B. On-air bytes per record, by placement:

$$A \text{ (inline)} : s + g_a + H_f/b, \quad (1)$$

$$B \text{ (self-batch)} : s + (g_a + H_f)/b, \quad (2)$$

$$D \text{ (block, } n \text{ frames)} : s + (g_a + nH_f)/b. \quad (3)$$

A. T1: overhead dominance

For inline authentication the pure-authentication fraction is

$$\varphi = \frac{g}{s+g}. \quad (4)$$

For $s = 45\text{--}190\text{ B}$ and $g = 64\text{ B}$, $\varphi = 25\text{--}59\%$: the smaller the payload, the more the signature dominates. This is the disease; T2–T5 are the cure and its limits.

B. T2: batching cure and amplification

Self-batch folds b records under one signature, so the per-record auth cost $(g_a + H_f)/b \rightarrow 0$. Packing to the MTU gives a goodput amplification over single-record framing:

$$A = \frac{M}{M - H_f - g_a}, \quad (5)$$

since at the MTU-filling batch $b_{\max}$ the frame $\approx M$ and $b_{\max}s \approx M - H_f - g_a$. Inline (A) can *never* amortize its per-record signature (only H_f/b shrinks).

T2a: when does A apply? (*applied analysis, not a new result* — the diminishing-returns structure below is standard in the packet-aggregation literature.) The derivation assumes the MTU caps the batch. Under a freshness bound the cap is instead $b = \lfloor \Lambda D_{\max} \rfloor$, which does not depend on s ; then $C(s) = s + (g_a + H_f)/b$ and $dC/ds = 1$ *exactly*—compression pays $1\times$, not $A\times$, and the residual authentication cost is a floor compression cannot reach. The boundary is $s < (M - H_f - g_a)/(\lfloor \Lambda D_{\max} \rfloor + 1)$. At $\Lambda = 20\text{ rec/s}$ and $D_{\max} = 250\text{ ms}$ this is 232.7 B on 802.11 , above every encoding studied, so **A is never operative there**; on a low-rate link ($M = 222$) the MTU binds again and $A = 1.88$ is operative. The “compression pays $\times A$ ” leverage is therefore real and *exclusive to low-rate links*.

C. T3: loss-robustness frontier

Let p be per-frame loss and $V = \Pr[\text{a received record is independently verifiable}]$. Self-batch frames self-verify, so $V_B = 1 - p$, *independent of b* . A block signature spanning $n(b) = \lceil (bs + g_a + H_f)/M \rceil$ frames needs all fragments, so $V_D = (1 - p)^n$. Whenever $n > 1$, $V_B = 1 - p > V_D = (1 - p)^n$: self-batch Pareto-dominates the marginal byte saving of over-aggregation.

D. T4: power-independent scheme crossover

Per-record energy is $E = P_c$ (CPU time) + P_r (radio time). Comparing Ed25519 and BLS, which wins depends on the powers *only* through their ratio $\kappa = P_r/P_c$:

$$\text{BLS wins} \iff \kappa > \kappa^* = \frac{\Delta\text{CPU}}{\Delta\text{RADIO}}, \quad (6)$$

where ΔCPU is BLS’s extra CPU and ΔRADIO its radio-time saving. κ^* is the break-even power ratio, obtained without measuring a single watt. Wi-Fi receive power sits below CPU-active power, so plausible platforms have $\kappa \lesssim 0.5$.

E. T5: co-design optimum

Enumerate $(e, \sigma, \text{placement}, b)$ and take the Pareto front over $\{\min \text{ bytes, min energy, min staleness, max } V\}$ subject to MTU fit, $V \geq 1 - \epsilon$, freshness $D(b) \leq D_{\max}$, and verify-throughput $t_{\text{verify}}(b)\Lambda \leq 1$. The optimum is the smallest deterministic encoding under self-batch at the largest *freshness-feasible* batch — not the largest MTU-feasible one. Which ceiling binds is itself a result (T2a): at telemetry rates fill time dominates $D(b)$, so $b \lesssim \Lambda D_{\max}$ and the MTU is never reached.

F. The payload-exclusion bound (applied, not new)

T1–T5 ask what a saved byte buys. This bound asks the prior question — whether a link admits authenticated telemetry at all. The inequality is not ours: Gündoğan et al. [1] compute exactly this residual for $802.15.4/\text{NDN}$, and the post-quantum literature reports NIST signatures as incompatible with 5G SIB1’s 372 B limit. We restate it because it delimits our contribution, and because one composition of it appears not to have been made explicit (below). A frame that is independently verifiable must carry the header, the authentication object and at least one record, so a link of maximum payload M is usable iff

$$s_{\max}(M) = M - H_f - g_a \geq s_{\min}. \quad (7)$$

Fragmenting an oversized signature is no escape: by T3 a unit spanning n frames verifies with probability $(1 - p)^n$, so $V \geq 1 - \epsilon$ forces $n \leq \lceil \ln(1 - \epsilon)/\ln(1 - p) \rceil$, which is exactly 1 whenever $\epsilon \leq p$. Equation (7) is therefore evaluated at $n=1$ and is an exclusion, not an inconvenience. Both ingredients are individually known — the residual above, and that a fragmented unit needs every fragment — but the consequence is worth stating: the *sliced-signature* workaround proposed for constrained links is unavailable in any regime where the verifiability target is no looser than the loss rate.

Three tiers follow, and each names a different escape route: $M < g_a$ (only a smaller signature helps — no header design, encoding, batching or chaining can), $g_a \leq M < H_f + g_a$ (only a leaner header), and $H_f + g_a \leq M < H_f + g_a + s_{\min}$ (the sole regime an encoder can attack, and where T2’s amplification is operative). Section VI applies this to LoRa.

G. Channel and energy models

At the 6 Mb/s OFDM base rate, broadcast airtime is $T_{\text{air}}(L) = T_{\text{phy}} + 8(L + \text{MAC})/R + \text{DIFS} + \delta$ (fixed part $\approx 100\text{ }\mu\text{s}$); unicast adds SIFS and an ACK ($\approx 156\text{ }\mu\text{s}$). Under contention among N saturated stations, the Bianchi DCF fixed point is

$$\tau = \frac{2(1 - 2p_c)}{(1 - 2p_c)(W + 1) + p_c W (1 - (2p_c)^m)}, \quad p_c = 1 - (1 - \tau)^{N-1}, \quad (8)$$

with $W = 16$, $m = 6$, solved by damped iteration; saturation throughput $S = P_{tr} P_s 8L/E[\text{slot}]$. Self-batch per-record energy is $E = P_c(t_{\text{enc}} + 2t_{\text{hash}} + t_{\text{sign}}/b + t_{\text{verify}}/b) + P_r T_{\text{air}}(\text{frame})/b$. The hash term is the chain link $\text{prev_hash} = H(\text{record})$, charged twice per record because the sender

TABLE II
E1: OVERHEAD DOMINANCE (T1), 30 SEEDS. $\varphi = 64/(s + 64)$.

encoding	mean bytes s	φ
JSON	191.1	25.1%
CBOR	66.25	49.1%
MessagePack	65.16	49.6%
delta	45.00	58.7%

extends the chain and the receiver re-verifies it; unlike the signature it does *not* amortize over b .

IV. IMPLEMENTATION

The project ran in phases under an eight-law discipline (plan-first, verify-before-assume, never-bypass-a-failure, TDD, audit-attack-fix, check-results, scientific-integrity, escalate-decisions); `main` is always green and every result carries an environment and config-hash header. **P0** locked the toolchain and CI. **P1** built four encoders and three schemes under a strict timing harness (GC off, warmup, checksum, bootstrap CIs) validated against real KATs (RFC 8032, Wycheproof, Chia). **P1b** switched the binary codecs to canonical schema arrays (CBOR 110→66 B). **P2** built the hash-chained store and frozen canonical-CBOR wire vectors. **P3** implemented the four placements and the 802.11a airtime model (broadcast and unicast kept separate). **P4** turned T1–T3 into seeded experiments (E1–E3). **P5** built the energy model, the exhaustive Pareto optimizer, and the power-independent crossover (E4). **P6** validated the Bianchi model against NS-3 3.48. **E5** fed the optimizer measured inputs and tested the headline. **P7a** produced the hardware-measurement scripts; a whole-repo audit then corrected one inconsistency and added an automated frozen-data reproduction gate (Sec. VII).

V. RESULTS

All values are from frozen CSVs. Byte and model results were derived on an x86 host (Intel i5-14400F); the crypto timings, encode timings and both power figures feeding the energy column are *measured on the Raspberry Pi 4* thesis platform (decision D8), so the energy column is platform-consistent ARM throughout.

E1 (Table II). The auth fraction climbs from 25% (JSON) to 59% (delta): compressing the payload makes a fixed 64 B signature dominate—T1 confirmed.

P1 crypto (x86 medians). ECDSA-P256 sign/verify 26.2/78.5 μ s; Ed25519 29.1/95.0 μ s; BLS 324.6/1016.5 μ s; BLS aggregate-verify($b=8$) 3445 μ s. Notably, ECDSA *beats* Ed25519 on x86 (OpenSSL P-256 assembly)—the reverse of the usual ordering. The ordering *flips* on the ARM target (Ed25519 259.5 vs ECDSA 326.9 μ s verify), which is what the optimizer uses; both are 64 B, so the byte results are identical either way. This is the reason the optimizer breaks a byte-tie toward ECDSA on the energy metric; this edge is expected to vanish on ARM.

E2 (T2/T2a). At MTU 1500, self-batch, the measured amplification $A_{@b} = (\text{bytes/rec})/s$ matches $A = M/(M -$

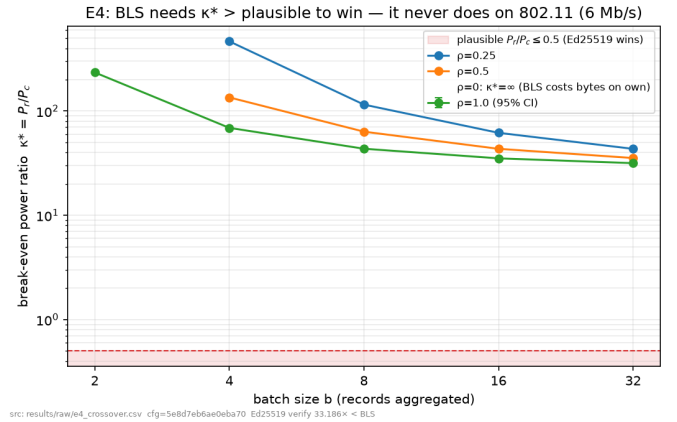


Fig. 1. E4: break-even power ratio $\kappa^*(b)$ per relay fraction. Every curve sits far above the plausible band ($\kappa \leq 0.5$), so Ed25519 wins on 802.11; the $\rho=0$ curve is ∞ (BLS costs bytes on own traffic).

TABLE III
E5: CO-DESIGN HEADLINE (T5) UNDER $V \geq 0.95$, $p=0.05$ AND $D \leq 250$ MS. TOTAL-BYTE CUT = $(174.3 - 72.0)/174.3 = 58.7\%$; AUTH CUT = $(108 - 27.0)/108 = 75.0\% \geq 40\% \Rightarrow$ PASS. $H_f=44$ B IS MEASURED FROM THE IMPLEMENTED WIRE FORMAT, NOT ASSUMED.

config	enc	σ	place	auth B/rec	total B/rec	V	D
optimized	delta	Ed	B, $b=4$	27.00	72.00	0.95	
A+CBOR (Pillar-1)	cbor	Ed	A	108.0	174.25	0.95	
A+JSON (naive)	json	Ed	A	108.0	299.09	0.95	
D-over-agg	cbor	Ed	D, $b=40$	3.80	70.05	0.9025	

$H_f - g_a$) within 5% (integer- b slack, shrinking with s); delta reaches $b_{\max} = 31$ with $A_{@b} = A = 1.0745$ exactly. The MTU sweep also straddles the T2a boundary: $M=256$ is MTU-limited ($A=1.68$ realised), while $M=576$ and $M=1500$ are freshness-limited and realise $A=1.0000$ —i.e. the $b_{\max}=31$ above is *unreachable* at telemetry rates.

E3 (T3). $V_B \approx 1 - p$ flat in b (0.90 at $p=0.10$); $V_D = (1 - p)^n$ drops the moment a block spans $n > 1$ frames (0.81 at $p=0.10$, $b=40$)—the frontier is exactly T3.

E4 (T4, Fig. 1). On ARM—the thesis platform—Ed25519 verify is $33.2\times$ cheaper than BLS verify ($10.7\times$ on x86). Across the full relay-fraction $\times$ batch $\times$ rate grid (80 points), Ed25519 wins *every* one (minimum $\kappa^* = 31.6 \gg 0.5$). At 96 B, BLS carries more auth bytes than Ed25519 on own self-batch traffic ($\kappa^* = \infty$) and saves bytes only on relay traffic for $b \geq 2$; its $33.2\times$ verify cost on ARM keeps Ed25519 optimal for any plausible power ratio.

E5 (Table III, Fig. 2). The optimizer selects delta + Ed25519 + self-batch at $b=4$, cutting total on-air bytes from 174.3 to **72.0 B/record**—a **58.7%** reduction versus the Pillar-1 baseline at $V = 0.95$, $p = 0.05$. Hand-checked: $45.0 + 108/4 = 72.0$ B/record. The same $b=4$ is reached at either operating point, with freshness $4/50 = 80$ ms at the adopted compliant point and $4/20 = 200$ ms at the relaxed one.

Feasibility is the co-design result (Table V). The substantive claim is not how many bytes are saved but which

TABLE IV

ATTRIBUTION OF THE E5 SAVING. THE AUTHENTICATION TERM IS $1 - 1/b$ AND IS INVARIANT TO H_f, g_a , ENCODING AND SCHEME; THE ENCODING TERM MOVES PAYLOAD ONLY. REPORTING THE SINGLE 75.0% FIGURE WOULD CREDIT FOUR AXES FOR WHAT TWO PRODUCE.

axis	moves	reduction	share of saving
placement $\times$ batching	auth bytes 108 $\rightarrow$ 27.0	75.0%	79.2%
encoding	payload bytes 66.25 $\rightarrow$ 45.0	32.1%	20.8%
scheme	neither (both 64 B)	—	—
combined	total bytes 174.3$\rightarrow$72.0	58.7%	100%

TABLE V

FEASIBILITY ACROSS THE OPERATING REGION, NOT AT ONE CHOSEN POINT (802.11A 6 Mb/s, BROADCAST MODEL OF [2]). $N_{\max}$ IS GIVEN AT BOTH THRESHOLDS: SATURATION ($U < 1$) AND THE MEASURED $V \geq 0.95$ BOUNDARY ($U \approx 2.80$). BATCH OBEYS $b \leq \Lambda D_{\max}$, SO **BYTES DEPEND ONLY ON THE PRODUCT** — THE FIRST TWO BLOCKS ARE BYTE-IDENTICAL AND DIFFER ONLY IN CAPACITY AND IN COMPLIANCE.

configuration	Λ (rec/s)	$D_{\max}$ (ms)	b $U < 1$	$N_{\max}$ $V \geq 0.95$
<i>Primary: compliant with 3GPP TS 22.125 [4]</i>				
optimized delta/B	50	100	4	35
A+CBOR (Pillar-1)	50	100	1	18
<i>Relaxed freshness: same bytes, larger swarm, exceeds the standard</i>				
optimized delta/B	20	250	4	103
A+CBOR (Pillar-1)	20	250	1	32
A+JSON (naive)	20	250	1	25
<i>Compliant but below the batching threshold ($\Lambda D_{\max} < 4$)</i>				
optimized delta/B	20	100	1	34
optimized delta/B	10	100	1	78

configurations *can run*. Coupling the byte model to the broadcast channel model gives the largest neighbourhood each configuration sustains. Two thresholds are reported because they answer different questions. *Saturation* throughput ($U < 1$) is what the medium carries when every node is backlogged, and it is conservative: simulation shows 98.8% delivery still at $U=1$, with *measured* verifiability V_{meas} crossing 0.95 only near $U \approx 2.44$. We report both rather than choose, because choosing would select the flattering one.

Across the operating region the co-design multiplies the supportable swarm by between $1.9\times$ and $3.3\times$ (Table V): $1.94\times$ and $3.31\times$ at the two thresholds for the compliant point, $3.22\times$ and $2.24\times$ for the relaxed one. **The advantage is robust in existence and variable in size**, and we state it that way; a single “ $\approx 3\times$ ” would be true of two of those four numbers and misleading about the other two. What no single axis produces is the advantage itself — it needs the encoding (frame size), the placement and batch (frames per second), the scheme (auth bytes) and the channel model together.

Choosing the operating point, and what it costs (item B3). Because $b \leq \Lambda D_{\max}$, *only the product $\Lambda D_{\max}$ sets the batch*, and therefore the bytes. Two very different operating points give the same $b=4$ and the same 72.0 B/record: (20 Hz, 250 ms) and (50 Hz, 100 ms). They are not equivalent, because 3GPP TS 22.125 [4] requires ≥ 10 msg/s and ≤ 100 ms latency, which the first violates. **We adopt the compliant**

point (50 Hz, 100 ms) as primary. It is not a contrived rate: 50 Hz is PX4’s MAVLINK_MODE_ONBOARD companion stream rate, read from `mavlink_main.cpp`. The byte result is unchanged; what it costs is swarm size, $233 \rightarrow 116$ at the $V \geq 0.95$ boundary and $103 \rightarrow 35$ at saturation. The relaxed point is retained in Table V because a deployment outside the standard’s scope may prefer it, and because reporting only the compliant point would hide the size of the trade.

The region also has a floor. Keeping $b \geq 4$ under a 100 ms bound needs $\Lambda \geq 40$ Hz; at 20 or 10 Hz the same deadline forces $b=1$ and the co-design collapses to a 12.2% saving. **The batching benefit is therefore not available at every compliant operating point** — it requires a telemetry rate high enough that four records accumulate within the deadline, which is a constraint on the application, not on the cryptography.

Attributing the saving (Table IV). We separate the axes rather than report one figure, because they are not equal contributors. Authentication overhead per record is $(H_f + g_a)/b$; the record size does not appear, so the auth-byte reduction $[(H_f + g_a) - (H_f + g_a)/b] / (H_f + g_a) = 1 - 1/b$ is invariant to header size, signature size, encoding and scheme. We verified this by substitution over $H_f \in \{20, 40, 80, 200\}$ B $\times g_a \in \{48, 64, 96\}$ B: all twelve give 75.0000%, and measuring H_f at 44 B rather than the assumed 40 B moved it not at all. Independently, the frozen E2 data gives the same auth overhead to three decimals for all four encodings. The encoding axis is nonetheless load-bearing—it spans $3\times$ on total bytes—but it acts on payload, contributing 20.8% of the saving against placement and batching’s 79.2%. The scheme axis is byte-neutral here (Ed25519 and ECDSA are both 64 B) and is decided on energy and verification throughput instead (T4).

Which constraint binds is itself the result. Ignoring freshness, the byte-optimal batch is $b=30$ at the MTU knee, cutting 96.7%—but the oldest record in such a batch waits **1.50 s**, $6.0\times$ over $D_{\max}$. Because fill time dominates the delay, the admissible batch obeys $b \lesssim \Lambda D_{\max}$ *independent of encoding and scheme*, so freshness binds long before the MTU does at telemetry rates. The co-design answer is therefore a frontier, not a point: $b=1 \rightarrow 4$ trades 0/50/66.7/75% auth-byte reduction against 50/100/150/200 ms of staleness, with sharply diminishing returns. Over-aggregation (D) is byte-competitive (3.60 B) but *fails* $V \geq 0.95$ ($0.9025 = (1-p)^2$)—a live demonstration of the T3 frontier—and misses freshness by $8\times$.

NS-3 validation (Table VI, Fig. 3). Unicast validates the Bianchi model [16] to $+1.28/-0.49\%$ across $N = 5-50$. Broadcast requires a *different* model, not the same one with the ACK removed. Reducing the unicast model by fixing $\tau = 2/(W+1)$ under-predicts NS-3 by $16\times$ at $N=50$. PHY-trace instrumentation shows the cause is not frame capture—which we measure at **0%**, every successful decode arriving from a single-transmitter busy period—but the *backoff counter consecutive freeze process*: with no ACK the contention window never doubles, so a station that has just transmitted may

E5 co-design (T5): optimized cuts auth bytes 75.0% vs A+CBOR at $V \geq 0.95$, $p=0.05$ [PASS]

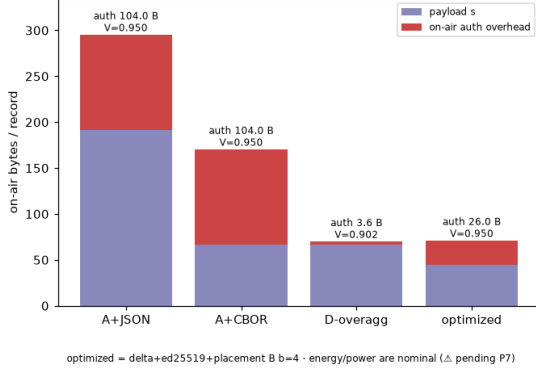


Fig. 2. E5: on-air bytes/record (payload + authentication) for the optimized configuration versus the baselines. Authentication overhead falls from 104 B to 26.0 B under the 250 ms freshness bound.

TABLE VI
ANALYTIC DCF MODELS VS. NS-3 3.48, RELATIVE GAP.

N	unicast vs. [16]	broadcast vs. naive reduction	broadcast vs. [3]
5	+0.6%	-0.2%	-0.3%
10	+0.9%	+3.1%	-0.4%
20	+0.6%	+31.3%	-0.7%
35	-1.4%	+273.1%	+0.9%
50	-2.9%	+1654%	+1.1%

redraw a zero backoff and seize the medium one slot ahead of every deferring station, whose counter is necessarily ≥ 1 . This is precisely the effect analysed by Ma and Chen [2], [3], whose closed form reproduces our measurements to within 2.49% on success probability, idle-slot occupancy and throughput; the unicast counterpart is the “anomalous slot” analysis of Tinnirello *et al.* [20], which our unicast traces independently confirm. We claim no novelty for the mechanism; our contribution is its validation at $W_0=16$ under 802.11a timing (the original studies used $W_0=32, 128$ at 1 Mb/s) and a direct slot-level measurement of it.

VI. GENERALISATION: THE LOW-RATE REGIME

Everything so far concerns a companion-class 802.11 link. The natural question is how far the co-design carries, and the honest answer is that it stops — for reasons that are worth stating precisely, because they bound the contribution rather than extend it.

This section is *analysis and simulation, not measurement*. No LoRa hardware was instrumented and no LoRa energy figure is reported: the only metered radio power in this work is a Wi-Fi figure, and reusing it would be fabrication. Every LoRa constant is transcribed from the SX1276 datasheet (Rev. 7, §4.1.1.5/§4.1.1.7) and LoRaWAN RP002-1.0.3 (Tables 8 and 13, EU868 duty cycle $< 1\%$).

A. T6 excludes the four longest-range modes

With $H_f = 44$ B and a 64 B Ed25519 signature, Eq. (7) partitions EU868 (Table VII). DR0–DR2 fail on the *signature alone*: 64 B does not fit a 51 B payload, so the exclusion

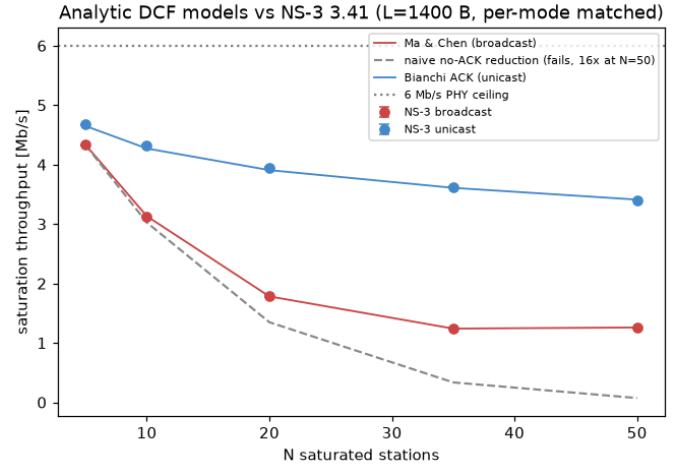


Fig. 3. Analytic DCF models vs. NS-3 3.48 saturation throughput. Unicast matches Bianchi within $+1.28/-0.49\%$; broadcast matches Ma and Chen [3] within 1.44%, while the naive reduction of the unicast model (dashed) fails by $16\times$ at $N=50$.

TABLE VII
T6 APPLIED TO EU868 ($H_f=44$ B MEASURED, $g_a=64$ B, $s_{\min}=13$ B).

DR	M (B)	$s_{\max}$	verdict
0, 1, 2	51	-57	excluded — signature alone overflows
3	115	7	excluded — no record fits (misses by 6 B)
4, 5, 6	242	134	feasible

holds even with a zero-byte header and a zero-byte record. The only escape is a smaller authentication object, and the smallest standardised one is a 48 B compressed BLS12-381 $\mathbb{G}_1$ point [21] at 126-bit security — which does fit 51 B, leaving three bytes for the header and the record together. DR3 fails on the encoding, and by six bytes.

B. Where the design changes: chaining

On 802.11 freshness binds, so the batch is fixed by $\Lambda D_{\max}$ and saving payload bytes buys nothing extra. On LoRa the *regional payload limit* binds, so bytes convert directly into more records per frame. Moving the 32 B chain link from per-record to once-per-frame — the receiver re-derives the rest from $prev_hash_{i+1} = H(\text{record}_i)$, leaving the stored ledger byte-identical — raises the DR5 batch from 2 to 7 and the sustainable rate from 0.060 to 0.182 rec/s, a factor of 3.03. We adopt it *on this arm only*: the same change on 802.11 buys $\approx 6\%$ energy and no extra records, which does not pay for losing independently transmitted per-record tamper evidence.

C. Capacity: the two penalties compound

Time on air and the duty cycle are deterministic, so simulating them would be circular. One quantity is not derivable that way — how many nodes share the channel — because LoRaWAN uplinks are pure ALOHA, with no carrier sense and no acknowledgement.

A topology caveat, stated because it differs from the 802.11 arm. LoRaWAN is by specification a star-of-stars

with no peer-to-peer mode: a recent FANET survey notes that it “presents some limitations regarding its star topology, its medium access control (MAC) layer and its lack of routing procedures” [22], and published peer-to-peer LoRa meshes are built by discarding the LoRaWAN MAC entirely, “not relying on LoRaWAN...without the use of gateways” [23]. A LoRa FANET that was actually flown reports the same constraint — “LoRaWAN only supports star topology” — and, having built one, converges on a *single channel* and on dedicated transmit and receive radios to work around half-duplex [24], which is the configuration we model. Nor is the gateway framing unrepresentative of the multi-hop literature: a review of mesh proposals for LoRa finds that even those pursuing flexible topologies retain “a predominant nodes → gateway/sink data flow”, with “fewer proposals” targeting genuinely decentralized architectures, and lists “network scalability with hundreds, or even thousands, of nodes” among the challenges that remain open [25]. Our scenario is therefore N end devices reporting to one gateway, whereas the 802.11 arm is a genuine broadcast among peers. **The two arms bracket the design space rather than repeating one experiment at two data rates**, and the low-rate result should be read as the infrastructure-collected variant. What transfers is the part that matters: collision statistics at a receiver depend on how many transmitters are concurrently in range, not on what the receiver is attached to, so the measurement models a single receiver in an ad hoc network with $N+1$ nodes. It is also why the single-demodulator provisioning is the right one here and not, as it would be for a gateway, an artificially harsh one: *a UAV peer has one radio*. Two ad hoc effects remain outside the model. A half-duplex radio cannot receive during its own 364 ms frame, costing a peer $\approx 2\%$ of every other peer’s frames, a loss no gateway suffers — and not an artifact of our setup, since Class A devices are specified as half-duplex pure-ALOHA transceivers [22] and peer-to-peer LoRa implementations report designing around exactly this constraint [23]. And V is a *per-receiver* metric, as in the 802.11 arm: a record reaching *every* peer in a single broadcast has probability $\approx (1-p)^{N-1}$, so single-hop full replication is not what V measures and would require gossip or relay — an application-layer mechanism we do not study. Simulating the AUTHBC frame at DR5 under the same 0.95 threshold used for 802.11 — here applied to *measured* delivery V_{meas} , not to the assumed V_{tgt} of the byte model — gives $N_{\text{max}} = 3$, with a sharp cliff: delivery is 0.960 at $N=3$ and 0.917 at $N=5$. There is no backoff to absorb contention.

This figure is far below the node counts usually quoted for LoRaWAN, and the discrepancy is configuration, not disagreement. We simulate the LoRa physical layer — real modulation, with the module’s measured interference and capture model — but on its harshest MAC provisioning: a single logical channel, a single gateway demodulation path, and one forced spreading factor, since the data rate is a design variable here and each run therefore fixes it. Comparison with measurement-based published results [26] must respect that. Both studies transmit at the 1% duty-cycle ceiling, so per-node

occupancy is 1% in both and offered load scales identically as $G = 0.01N$; mapping their multi-channel result through their own fit $f_{\text{MCH,MSF}}(x) = f_{\text{SCH,SSF}}(x/18)$ places their $\approx 32\%$ loss at 56 nodes on one channel and one SF, where we measure $\approx 75\%$ at $N=50$. **Our model is therefore roughly $2.3\times$ more pessimistic than theirs, not more optimistic**, most plausibly because a single demodulation path rejects a second concurrent arrival outright instead of granting it the capture chance their measurements support. Our own configuration is deliberately the pessimistic one in a further respect: the interference matrix treats *any* same-spreading-factor overlap as fatal, so there is no capture at all, where a real LoRa receiver recovers the stronger of two colliding frames. Enabling a 6 dB capture threshold raises delivery by 3.3 points at $N=8$ and by $1.36\times$ at $N=50$ without moving N_{max} . The small margin over the Poisson curve e^{-2G} at low load is not capture but traffic regularity: sources are periodic rather than Poisson, for which the escape probability is $(1 - 2\tau/T)^{N-1} = 0.868$ at $N=8$ against 0.866 measured.

The result is also range-limited, and that limit binds before capacity does. Our channel model attributes all loss to collisions, and this is not an option we simply left switched off: enabling correlated shadowing changes the outcome on no row at either 500 or 1000 m. That is a property of the link budget, not a wiring error — with 14 dBm transmit power and the -130 dBm SF7 gateway sensitivity, log-distance path loss leaves 28.8 dB of margin at 500 m and 17.5 dB at 1000 m, which an ≈ 8 dB shadowing standard deviation cannot erode; simulated failure onset is near 4200 m, and shadowing does bite there (1.000 falls to 0.515 at 4000 m). *Our propagation parameters describe a considerably better link than a real drone-to-drone channel, and no option the scenario exposes closes that gap.* Field measurements of air-to-air LoRa on real drones [24] show a link losing frames to range alone: PDR falls from 0.9711 at 200 m to 0.9550 at 500 m and 0.9045 at 1000 m, measured with two drones and a base station, where contention is negligible. Since the two loss mechanisms are independent, delivery is $P_{\text{link}}(d) \times P_{\text{no-coll}}(N)$, and any range whose link PDR already misses the target makes it unreachable at *every* node count. Composing their measured link with our measured collision curve is sharper than either term alone:

V target	200 m	500 m	1000 m
0.95	1	1	0
0.90	3	3	1
0.85	5	5	3
0.80	10	8	5

At $V \geq 0.95$ the regime admits no multi-node network at all. The link alone delivers 0.9711 at 200 m, leaving 0.0039 of headroom, while our least-loaded measured point ($N=2$) already spends 0.0283 on collisions. A lone transmitter passes; two do not. The capacity figure is therefore conditional not only on range but on the threshold: relaxing to $V \geq 0.90$ restores $N_{\text{max}} = 3$, and to 0.85 restores 5. We report the curve rather than a point because the choice of threshold, not the physics, is what decides whether this regime is usable

— and that choice belongs to the deployment, not to us. At the 1000 m radius our own scenario configures, an idealised channel is 9.6 percentage points optimistic; this mirrors the 802.11 arm, where the idealised model is 39% optimistic at 500 m.

$N_{\max} = 3$ **should therefore be read as a worst-case bound at short range — one channel, one demodulator, one spreading factor, $d \leq 500$ m — rather than as a LoRaWAN network capacity. It also carries wide uncertainty.** Every device transmits on one exact period and ALOHA has no backoff, so relative phases are fixed for a whole run: a pair that collides once collides always. Delivery is consequently bimodal rather than noisy about a mean — over 30 seeds at $N=5$, 22 runs deliver exactly 1.000 and the rest fall to 0.20–0.84, giving a mean of 0.891 with $\sigma = 0.21$. A three-seed mean, which is what the figure above rests on, cannot characterise that distribution. At realistic ± 20 ppm crystal tolerance the accumulated drift over an hour (144 ms) is too small to break a 728 ms collision window, so the effect is physical for a naive periodic sender; real LoRaWAN Class A devices randomise transmission timing precisely to avoid it, which the simulated sender does not model. We report the figure with this caveat rather than restate it as a precise capacity. It is conservative in the direction that matters least for the claim being made: the point of this section is that the low-rate regime is qualitatively different, and a conservative bound does not manufacture that difference, it understates the margin. It is conservative in one further respect: LoRaWAN Class A specifies pure ALOHA [22] and we model it, whereas a purpose-built LoRa FANET adds listen-before-talk [24] and would collide less.

D. What infrastructure collection would buy

The single-demodulator, single-channel provisioning above models a UAV peer. The complementary question — what if a ground gateway collected the ledger instead? — is a configuration change, and we ran it: three channels and eight demodulation paths, the RP002 provisioning, with the same forced data rate. Delivery improves substantially and the improvement grows with load, reaching $2.68\times$ at $N=50$ ($0.2532 \rightarrow 0.6781$) and still 0.5308 at $N=100$, which the peer configuration cannot reach at all. **Yet $N_{\max}$ moves only from 5 to 8.** The $V \geq 0.95$ criterion sits on a near-vertical part of the delivery curve — the gateway passes at $N=8$ with 0.9958 and fails at $N=10$ with 0.9370 — so a threefold reduction in offered load per channel shifts the threshold barely at all. This is the same duality the 802.11 arm shows between saturation and the measured verifiability boundary: *a capacity metric and a strict-reliability metric respond very differently to the same change*, and reporting either alone would mislead.

E. An external check on the low-rate capacity

The 802.11 arm is validated against two published analytical models. The low-rate arm rested on a single simulator, so we gave it the same treatment. Bor *et al.* [26] publish their capacity result in closed form — a quintic fitted at $R^2 = 0.997$

TABLE VIII
THE LOW-RATE CAPACITY, BY TWO INDEPENDENT METHODS.

N	AUTHBC (NS-3)	Bor <i>et al.</i>	
5	0.0%	5.1%	we are more optimistic
8	13.4%	7.0%	crossover
30	59.0%	19.9%	we are $3.0\times$ more pessimistic
50	74.7%	29.9%	$2.5\times$
$N_{\max}$ at $V \geq 0.95$			
	5	4	

TABLE IX
THE LOW-RATE REGIME IS NOT A SLOWER VERSION OF THE SAME PROBLEM.

	per node	$N_{\max}$	aggregate
802.11a, delta/B, $b=4$	20 rec/s	103	2060 rec/s
LoRa EU868 DR5, $b=6$	0.165 rec/s	5	0.82 rec/s
ratio	$121\times$	$21\times$	$\approx 2500\times$

to *total* packet loss (collisions plus wrong-payload-CRC) measured on SX1301 hardware — which means it can be *evaluated at our operating point* rather than quoted at theirs. We implemented it, validated the implementation against the four loss figures their own text states (three land within ≈ 3 points; the fourth is stated as collisions-only, so our total correctly exceeds it), and applied the identical $V \geq 0.95$ criterion.

A discrete-event simulation of the LoRa physical layer and a polynomial fitted to hardware interference measurements land **one node apart** on the quantity this section rests on (Table VIII). The agreement should not be over-read — below $N \approx 10$ their fit is dominated by a 1.78% intercept that is an artifact rather than a measurement — but it is obtained independently and it is the right order. The disagreement in *shape* is the more informative part, and it matches the configuration difference: our runs model no capture at all and a single demodulation path, where their measurement-based model grants both, so above the crossover we reject concurrent arrivals that their gateway would still recover.

The arm is $\approx 120\times$ slower per node *and* $\approx 20\times$ smaller per domain, and it is the product that matters (Table IX). LoRa is not a slow 802.11; it is a different regime, in which authenticated per-record telemetry is not the right application at all. Its defensible role is long-range *provenance* — a tamper-evident trickle — not live telemetry.

One caveat of provenance. The simulator enforces RP002 Table 12 (repeater-compatible, 222 B) while our analytical model uses Table 13 (non-repeater, 242 B); at DR5 that caps the batch at 6 rather than 7. Both are defensible readings of the same standard, and we simulate what the implementation accepts rather than reconcile them silently.

VII. REPRODUCIBILITY AND SCIENTIFIC INTEGRITY

Beyond the results, the project’s second product is its discipline. Anomalies were root-caused, not patched over: a

stateful delta codec used statelessly (60→45 B); string-keyed CBOR replaced by canonical arrays (110→66 B); the BLS size corrected 48→96 B (with the T4 model updated); and and, on the broadcast channel, three successive explanations retracted in writing—an “18× capture” over-claim, then the capture effect itself (PHY instrumentation measured it at 0%), and finally our own claim to have discovered the responsible mechanism, which a literature check showed had been published in 2007 [2]. A whole-repo audit re-derived every formula against the code and hand-checked one point per experiment. Finally, freezing raw data risks *silent staleness*, so an automated gate re-derives every deterministic frozen artifact from the current code and fails on any drift—verified to catch the exact regression it was built for. No result is fabricated, no test skipped, no tolerance widened.

VIII. LIMITATIONS AND FUTURE WORK

Results are 802.11-only (a LoRa arm is future work). Energy is now measured on a Raspberry Pi 4 with INA219 meters ($P_{\text{cpu}} = 0.634 \text{ W}$, $P_{\text{radio}} = 0.218 \text{ W}$ —both nominal values were 3–5× too high), and the byte-based headline is power-free either way. The energy model is *composed* from separately measured powers and timings, so we metered the composition end-to-end on four configurations. It was initially $\approx 32\%$ low; both causes were identified and corrected—an omitted per-record chain-hash term (SHA-256, 2745.5 ns measured on ARM, charged twice per record and not amortized over b), and a P_{cpu} measured over *isolated* primitives (0.634 W) that understates any composed pipeline by 18.2% (four configurations meter 0.732–0.760 W, a 3.8% spread, so a single constant remains appropriate; we adopt 0.749 W). The residual is +7.5 to +14.3% and is entirely frame assembly between crypto calls, which we deliberately do not charge because it is interpreter overhead of a prototype rather than a property of the design. Absolute energy figures should therefore be read as lower bounds by ≈ 10 –14%; byte results are power-free. The channel models are validated only in the idealised single-collision-domain regime they describe: a spread, mobile formation introduces spatial reuse and, at long range, hidden terminals that neither model represents. We bound this: sweeping a realistic air-to-air scenario (Friis, 3D cluster, Nakagami fading) shows the idealised model is a *conservative* lower bound up to a ≈ 300 –400 m cluster (goodput +42 to +51% from spatial reuse) but becomes *optimistic* by 39% at 500 m, where links break down; Rayleigh fading ($m=1$) costs a further 27 points. These two marginal-link figures moved substantially when we migrated NS-3 3.41→3.48 (from 15.7% and 9 points), while every near-field scenario moved under 2% — consistent with that release’s InterferenceHelper and WifiPhy fixes, which bite hardest at marginal SNR. We report the newer simulator’s figures. The hardware validation is two-node. On ARM the scheme tie breaks toward Ed25519 rather than ECDSA, with an identical auth-byte cut.

IX. CONCLUSION

Authentication overhead dominates tiny UAV telemetry, but it is a *co-design* problem. Jointly optimizing encoding, placement, scheme, and batching cuts total on-air bytes by 58.7% at $V \geq 0.95$ within the 100 ms freshness bound of 3GPP TS 22.125—meeting the falsifiable criterion at a standards-compliant operating point.

We are deliberate about where that result comes from. The authentication component, taken alone, reduces to $1 - 1/b$: the header and signature sizes cancel, and neither the encoding nor the scheme appears. Reported as a headline percentage it would overstate what four coupled axes contribute. The contribution that genuinely requires the coupling is *feasibility*—which configurations can run at all. Joining the byte model to the channel model shows the inline baselines exhaust a single collision domain at 25–32 neighbours while the co-designed configuration reaches 103 — a $\approx 3\times$ ratio that is stable under either feasibility threshold.

Pushed to its limit, the same coupling yields an exclusion result that we regard as the more durable contribution. A link whose payload cannot hold one header, one signature and one record carries no per-frame-verifiable telemetry at *any* encoding or batch size, which removes the four longest-range LoRa modes outright; loss makes fragmentation no escape, since a unit spanning n frames verifies with probability $(1 - p)^n$ and the target is already spent at $n=1$ whenever $\epsilon \leq p$. It is not escapable by better cryptography, and it is not visible to a byte model alone. Capacity, by contrast, turned out to constrain rather than exclude: simulation shows the channel still delivering 98.8% of frames at nominal saturation load, so adopting the 100 ms bound of 3GPP TS 22.125 [4] costs swarm size— $N \leq 116$ instead of $N \leq 233$ —and nothing in bytes, provided the telemetry rate is high enough for the batch to fill inside the deadline. An NS-3-validated channel model and an automated reproducibility gate keep the study honest and repeatable.

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